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# All-metal metamaterial slow-wave structure for high-power sources with high efficiency

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In this paper, we have proposed a metamaterial (MTM) which is suitable for the compact high-power vacuum electron devices. For example, an S-band slow-wave structure (SWS) based on the all-metal MTMs has been studied by both simulation and experiment. The results show that this MTM SWS is very helpful to miniaturize the high-power vacuum electron devices and largely improve the output power and the electronic efficiency. The simulation model of an S-band MTM backward wave oscillator (BWO) is built, and the particle-in-cell simulated results are presented here: a 2.454 GHz signal is generated and its peak output power is 4.0 MW with a higher electronic efficiency of 31.5% relative to the conventional BWOs. © *2015 AIP Publishing LLC*.
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The concept of left-handed metamaterials (MTMs) was introduced by Veselago as early as in 1968[1] and first realized until in 2000.[2] The MTMs have many unique electromagnetic properties that cannot be achieved in the natural materials, such as negative refractive index, reversed Cherenkov radiation (RCR), reversed Doppler effect, and so on.[1] Researchers around the world show intense interests in the realization of different types of MTMs and their potential applications at microwave, millimeter wave, terahertz (THz), and optical bands, such as an electromagnetic cloak and a super-lens. Specially, one of current research hot topics is the MTM's application to the vacuum electron devices[3–6] based on the in-depth studies of the RCR,[7,8] particularly RCR in a waveguide with double-negative MTMs.[9,10]

High-power vacuum electron devices such as traveling-wave tubes (TWTs) and backward wave oscillators (BWOs) have been widely developed for decades. Motivated by urgent demands in the various areas such as civil and military fields, in particular, the military field, the research on this relatively matured subject has been done continuously.[11,12] At present, main serious challenges with the high-power vacuum electron devices are from the miniaturization and high electronic efficiency of the devices. Generally speaking, the slow-wave structure (SWS) greatly determines the performance of the vacuum electron devices. The MTM SWS has important advantages such as the miniaturization and high interaction impedance due to the local strong axial electric field, which largely benefit the high power vacuum electron oscillators and amplifiers with high efficiency. In recent years, the fundamental physics of the beam-wave interaction in MTMs has been studied.[13–16]

The realizable MTM-loaded waveguide was first studied as early as 2002.[17] Marqués and Esteban *et al*. described that a MTM-loaded waveguide can support the quasi-TE or TM mode wave below the cutoff frequency of the empty metal waveguide, respectively.[17,18] This behavior is beneficial to the miniaturization of the waveguide.[19] Of course, the traditional coupled-cavity (CC) SWS also has the similar feature.[20,21] In addition, a study of the propagation characteristics of MTM-lined circular waveguides has been reported.[22] These unique features can provide the possibility for applying the MTM SWS to the vacuum electron devices and accelerators. In this paper, we present a MTM unit cell,[23,24] which can be used to construct a SWS for the coherent high-power vacuum electron devices, as shown in Fig. 1.

O'Brien and Pendry proposed a nanostructured photonic crystal composed of a circular metallic resonator.[25] This resonator has a strongly dispersive magnetic permeability due to the large internal inductance and capacitance of the structure. Based on the Babinet's principle, a MTM unit cell has been creatively developed as shown in Fig. 1. This MTM cell has a strongly dispersive electrical response. As a result, a strong local electronic field is formed in the central area of the structure, which can be utilized to effectively interact

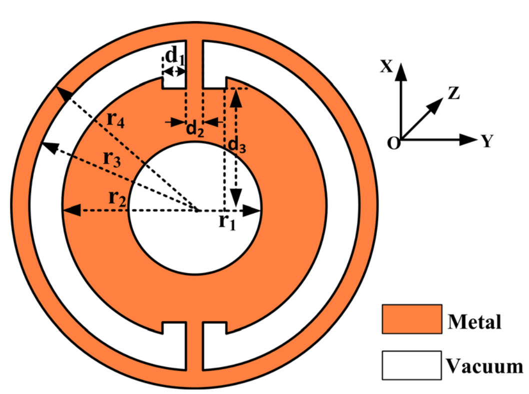


FIG. 1. Schematic of a MTM unit cell.

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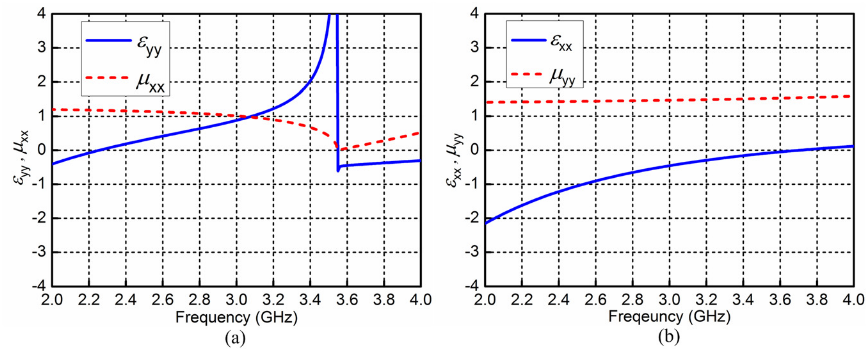



FIG. 2. Retrieved permittivities and permeabilities of the MTM as functions of frequency for $y$ (a) and $x$ (b) polarizations, respectively.

with the electron beam. The dimensions of the unit cell are set as $r_1 = 6$ mm, $r_2 = 15$ mm, $r_3 = 18$ mm, $r_4 = 20$ mm, $d_1 = 3$ mm, $d_2 = 2$ mm, $d_3 = 13$ mm, $p = 30$ mm (the period of the MTM unit cells), and thickness $t = 1$ mm (not shown here) in order to operate at frequencies from 2.2 to 2.6 GHz. Due to the fact that the frequency of the proposed MTM SWS is ~2.45 GHz, which means that $\lambda_0/p$ ($\lambda_0$ is the wavelength in free space) is about 4.08, it can be considered as an effective medium.[26,27] Then, we use the S parameter-based retrieval method[26] to retrieve the effective permittivity $\varepsilon_{yy}$ and $\mu_{xx}$ as well as $\varepsilon_{xx}$ and $\mu_{yy}$, as shown in Figs. 2(a) and 2(b), respectively. The permittivity tensor and permeability are determined for the waves propagating along the $z$-direction. Thus, the effective permittivity is anisotropic and the effective permeability is isotropic. The results are similar to those in the paper.[28]

Then, a SWS based on the MTM unit cells has been proposed as shown in Fig. 3, the MTM SWS consists of a closed metallic circular waveguide loaded with the MTM unit cells periodically with the period $p$. The pencil electron beam tunnel with the radius 6 mm locates in the center axis of the MTM SWS. Here, the inner radius of the circular waveguide is 20 mm, thus the MTM SWS can be considered as one-dimensional MTM for the TM-dominant mode at frequencies 2.45–2.53 GHz (as shown in Fig. 6(a)), while the cross-sectional area of the SWS of a S-band CCTWT is between $25 \times 25\,\mathrm{mm}^2$ and $30 \times 30\,\mathrm{mm}^2$, so this fact means there is a smaller dimension at the same operating frequency for this proposed MTM SWS, which is very helpful to miniaturize the devices.[27]

In fact, there are pivotal differences between this MTM SWS and the traditional CCSWS. The axial electric field of the MTM SWS is enhanced by the strongly electrical response of the MTM which has no need of a drift tube, in contrast to the CCSWS. Based on the MTM theory, the grooves in the MTM unit cell mainly determine the frequency range in which the effective permittivity is negative. Meanwhile, the $n = 0$ space harmonic of the fundamental mode (see Fig. 6(a)) for the MTM SWS would only be a "backward" wave, and the interaction impedance of the fundamental mode is more larger than the higher-order modes, these features do not exist in the conventional CCSWS. More detailed discussions have been described.[27]

In order to verify the transmission characteristics of the MTM SWS, both simulation and experiment have been carried out as shown in Fig. 4.[27] Here, the MTM SWS consists of 11 unit cells and the corresponding size parameters are structured, as previously mentioned. A vector network analyzer Agilent N5230A PNA-L is used as the signal generator and receiver in the experiment, the signal is fed and received by two SMA coaxial lines to the circular waveguide junctions[15] (the radius of the junction is 50 mm, the optimized lengths of the input coaxial line and the junction are $L_1 = 100$ mm and $L_2 = 16$ mm). In addition, it needs to be pointed out that the signaling system is symmetric about the center of the MTM SWS. The results simulated by CST Studio Suite 2012 indicate that a TM-dominant mode electromagnetic wave can be propagated at certain frequencies that are below the cutoff frequency of the empty circular waveguide. Fig. 4 illustrates the simulated results and the measured results of the S-parameters for comparison, which clearly shows that the measured results are in good consistent with the simulated results. It can be observed that the operating frequency range of the MTM SWS is from 2.40 to 2.48 GHz which are below the cutoff frequency of the TM-dominant mode for the circular waveguide, i.e., 5.7 GHz. We also find that the simulated results have lower

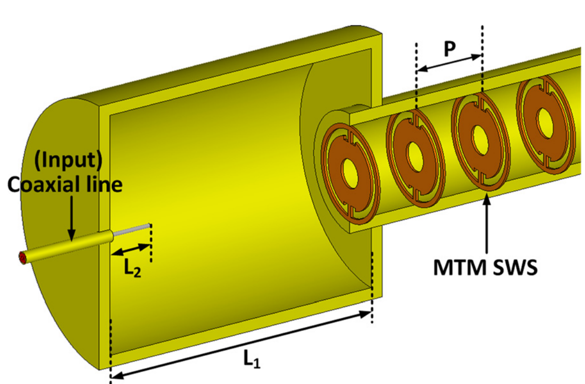



FIG. 3. Sketch of the experimental setup for the transmission characteristics of the MTM SWS.

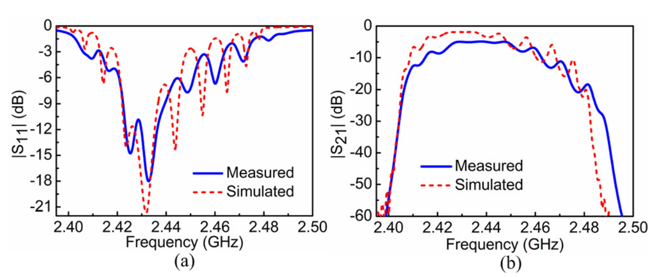



FIG. 4. Comparison between simulated and measured results of S parameters for the MTM SWS.

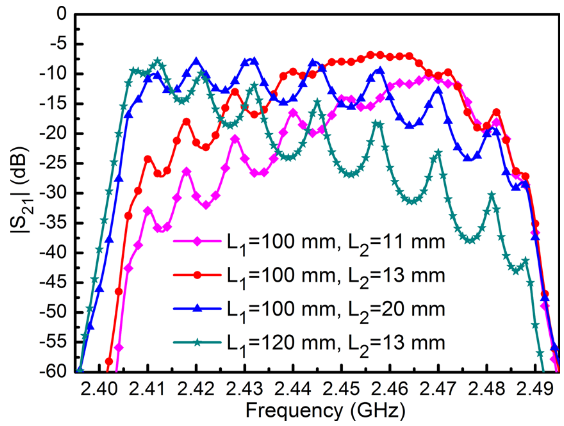



FIG. 5. Measured transmission coefficient ($|S_{21}|$) results with different lengths of $L_1$ and $L_2$.

transmission loss compared to the measured results. It is because the machining and assembling tolerance are inevitable in the experiment, such as the gaps between the components and the alignment of the MTM SWS.

Furthermore, we study the effects of the parameters of the signaling system on the transmission performance of the MTM SWS. The simulated results fully reveal that changing lengths of the circular waveguide junction and the coaxial line has a significant effect on the transmission performance. As shown in Fig. 5, the measured results of $S_{21}$ with changing $L_1$ and $L_2$ verify the point.

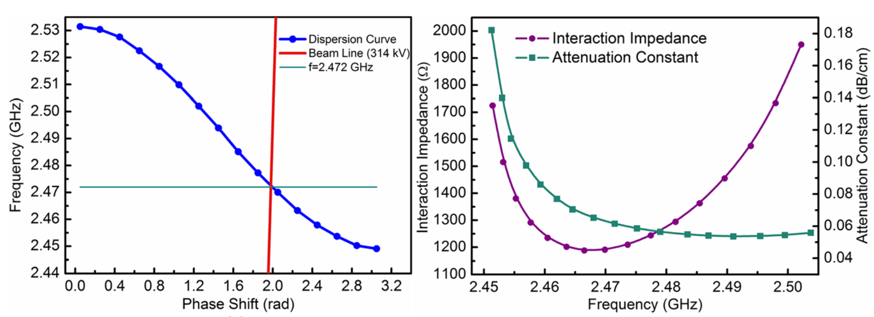
As a practical metallic structure, this MTM SWS can easily hold a vacuum environment and bear the high temperatures, it has a great potential to be used in the vacuum electron devices. Therefore, the high-frequency characteristics of the MTM SWS are studied by Ansoft HFSS 13.0 with master and slave boundaries,[29] as shown in Fig. 6.

Fig. 6(a) exhibits the dispersion characteristics of the MTM SWS from the HFSS simulations, it is found that the $n=0$ spatial harmonic (corresponding to the region from 0 to $\pi$) of the fundamental mode (the operating mode) is a "backward" wave, the frequency band of which is about from 2.45 to 2.53 GHz. The simulated results on the electric and magnetic fields also indicate that the fundamental mode is a TM-like mode wave. For the SWS, the interaction impedance of the $n$-th spatial harmonic is defined by: $\boldsymbol{k_{cn}} = |\boldsymbol{E_{zn}}|^2/2\boldsymbol{\beta}_n^2\boldsymbol{P_\omega}$, where $\boldsymbol{P_\omega}$ represents the transmission power along the axial direction, $\boldsymbol{E_{zn}}$ denotes the amplitude of the axial component of the electric field, and $\boldsymbol{\beta_n}$ refers to the phase constant of the $n$th spatial harmonic. Here, the MTM SWS operates on the zeroth spatial harmonic. The interaction impedance is the critical factor of judging a high power output for the vacuum electron devices. Generally speaking, the interaction impedance of the S-band helix SWS is from 100 to 200 Ω, and the S-band CCSWS has an interaction impedance between 300 and 400 Ω. Fig. 6(b) demonstrates the interaction impedance of the MTM SWS of the fundamental mode as a function of frequency. It is greater than 1150 Ω, which shows a largely improvement compared to the conventional SWSs. In addition, the interaction impedance of the first higher-order mode is lower than 1 Ω in the frequency band (from 3.2 to 3.7 GHz). Therefore, it is an important advantage for the MTM SWS that the risk of self-oscillation caused by the higher modes for the proposed devices can be ignored compared to the traditional CCSWS. In Fig. 6(b), the MTM SWS also shows a great performance of lower attenuation, which is conductive to increasing the output power and maintain the stability of the devices during the operation. Here, the attenuation constant is defined as $\boldsymbol{\alpha} = \boldsymbol{P_{loss}}/\boldsymbol{2P}$, where $\boldsymbol{P_{loss}}$ refers to the loss power, $P$ represents the power flow along the axial direction.

The analyses of the calculated results of high-frequency characteristics show that the MTM SWS is very suitable for the narrow band, coherent high-power vacuum electron devices. The all-metal and easily installed MTM SWS ensures that the devices have the advantages of better heat dissipation and easier fabrication. Subsequently, an S-band MTM BWO has been designed and then optimized by using the PIC solver in CST Particle Studio.[27] Based on the previous calculated results, the MTM SWS of the proposed S-band MTM BWO is composed of 15 MTM unit cells, and the whole length of the tube is about 510 mm. A pencil electron beam with a radius of 3 mm and a current of 40 A is assumed in the simulation, and its accelerating voltage is 314 kV which is determined by the operating frequency. Because the electron beam moves along the center axis of the SWS, the coaxial line adopted in the previous experiments is replaced by a coaxial coupling loop to form a beam tunnel as shown in Fig. 7. Additionally, a uniform longitudinal magnetic field of 0.5 T is used to focus the pencil electron beam. An output signal is produced at 2.454 GHz, and its peak power is 4.0 MW with an electron efficiency of 31.5%, which is bigger than the efficiency of the conventional BWOs (1%–15%).[30,31]

It should be noticed that the signal frequency of 2.454 GHz from the PIC simulations is slightly lower than 2.472 GHz which is determined by the cross point between the dispersion curve and the beam line.






(a) (b)

FIG. 6. (a) Dispersion curve of the fundamental mode for the MTM SWS. (b) Interaction impedance curve defined on the central axis of the beam tunnel, and attenuation constant curve of the MTM SWS.

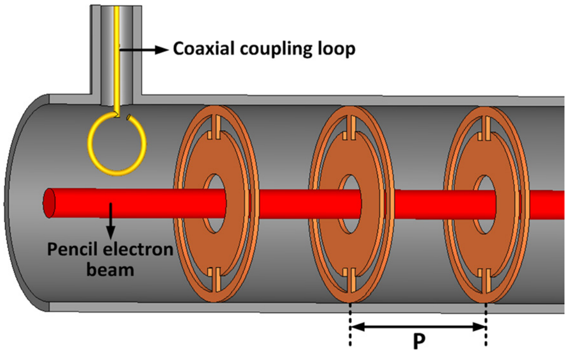



FIG. 7. Sketch of the PIC simulation model.

In conclusion, we have proposed a MTM unit cell and structured the size parameters in the S-band. And then the miniaturized all-metal MTM SWS is developed. The good agreement between the experiment and the simulation results demonstrates that the MTM SWS is very helpful to miniaturize the devices. The calculated results of the high-frequency characteristics also predict that this MTM SWS is suitable for the high output power vacuum electron devices with higher electronic efficiency, accelerators, and particle detectors. In the end, for instance, an S-band MTM BWO based on the SWS has been proposed and then calculated with the CST Particle Studio 2012. The calculated results show that it is a promising miniaturized, high-power microwave source. It has potential applications such as in narrow band communications and scientific equipment such as accelerators as well as industrial heating. In the future, we will carry out the related experimental work.

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